Waves at boundaries

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe reflection and transmission at a boundary.

SUCCESS CRITERIA - I can...

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A reflected signal returns after 3 s. Name the extra quantity needed to calculate distance and explain why this is a round trip.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

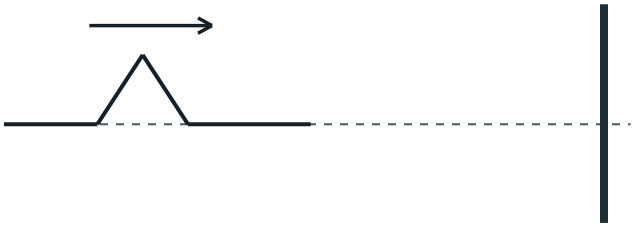
Worked reasoning: At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. These are different boundary conditions.

Your turn: Draw incident and reflected pulse snapshots for both ends; show propagation directions.

Marzano check - circle: ① ② ③

03 Look and explain

An upward rope pulse reaches an ideal fixed end. Sketch the reflected pulse and its travel direction.



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Build the solution

A supplied energy split gives 10 J incident, 4 J reflected and no other loss. Find transmitted energy.

Marzano check - circle: ① ② ③

05 Use the idea

Does a smaller transmitted amplitude prove frequency changed? Explain what usually fixes the frequency at a stationary boundary.

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg sample completely changes phase with latent heat 600 J/kg. Find the energy magnitude required.

Marzano check - circle: 1 2 3

07 Exit check

For an ideal free-end rope reflection, is an upward displacement pulse returned upright or inverted?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two equal upward pulses approach each other on an ideal rope. Sketch overlap and later separation. Explain whether they permanently stick.

Marzano check - circle: 1 2 3